

Central Limit Theorem

Statistics II

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Additivity (Normal Distribution)

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If:

1. $X_i \sim \text{Normal}(\mu, \sigma)$, $i = 1, \dots, n$ — independent and identically distributed random variables
2. $T = X_1 + \dots + X_n$ and $\bar{X} = \frac{T}{n}$

then:

$$T \sim \text{Normal}(\mu_T = n \times \mu, \sigma_T = \sqrt{n} \times \sigma)$$

$$\bar{X} \sim \text{Normal}\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$$

Example 1

Consider:

1. Independent variables: $X_i \sim \text{Normal}(\mu_i = 40, \sigma_i = 10)$

X_i — lunch duration of student i (in minutes)

2. $\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$ — sample mean ($n = 25$ and $n = 81$ students)

Find the probability that the sample mean lies between 39 and 41 minutes.

Solution: (assuming Normal distribution)

$$\bar{X} \sim \text{Normal}\left(\mu_{\bar{X}} = 40, \sigma_{\bar{X}} = \frac{10}{\sqrt{n}}\right)$$

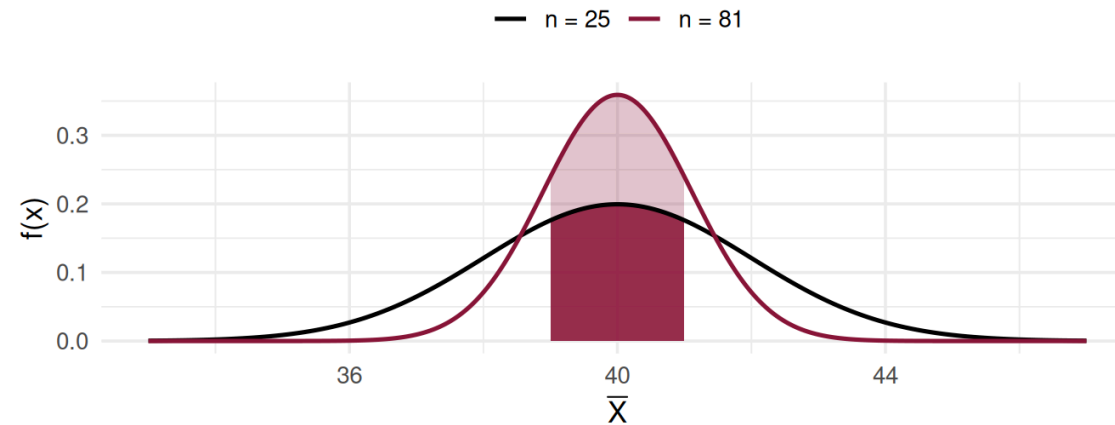
$$\Rightarrow P(39 < \bar{X} \leq 41) = P\left(\frac{39 - 40}{\frac{10}{\sqrt{n}}} < Z \leq \frac{41 - 40}{\frac{10}{\sqrt{n}}}\right)$$

Example 1 — Solution (cont.)

$$P(39 < \bar{X} \leq 41) =$$

- $n = 25$: $= P(-0.5 < Z \leq 0.5) = \Phi(0.5) - \Phi(-0.5) = \mathbf{0.3829}$
- $n = 81$: $= P(-0.9 < Z \leq 0.9) = \Phi(0.9) - \Phi(-0.9) = \mathbf{0.6319}$

Effect of Sample Size



Comment: As sample size increases, the probability that the sample observations concentrate around the mean also increases.

Central Limit Theorem (CLT)

Central Limit Theorem (CLT):

If:

1. $X_i \sim \text{Distribution}(\mu, \sigma)$, $i = 1, \dots, n$ — **independent and identically distributed random variables**
2. $T = X_1 + \dots + X_n$, $\bar{X} = \frac{T}{n}$

then, for n sufficiently large (as a rule, $n \geq 30$):

$$T \sim \text{Normal}(\mu_T = n \times \mu, \sigma_T = \sqrt{n} \times \sigma)$$

$$\bar{X} \sim \text{Normal}\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$$

Example 1 — Revisited (CLT)

Consider:

1. Independent variables: $X_i \sim \text{Distribution}(\mu_i = 40, \sigma_i = 10)$

X_i — lunch duration of student i (in minutes)

2. $\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$ — sample mean

$n = 25, n = 81, n = 225$ **students**

Find $P(39 < \bar{X} \leq 41)$. Solution: (without the assumption of normality)

With **$n \geq 30$** :

$$\bar{X} \sim \text{Normal}\left(\mu_{\bar{X}} = 40, \sigma_{\bar{X}} = \frac{10}{\sqrt{n}}\right) \xrightarrow{CLT}$$

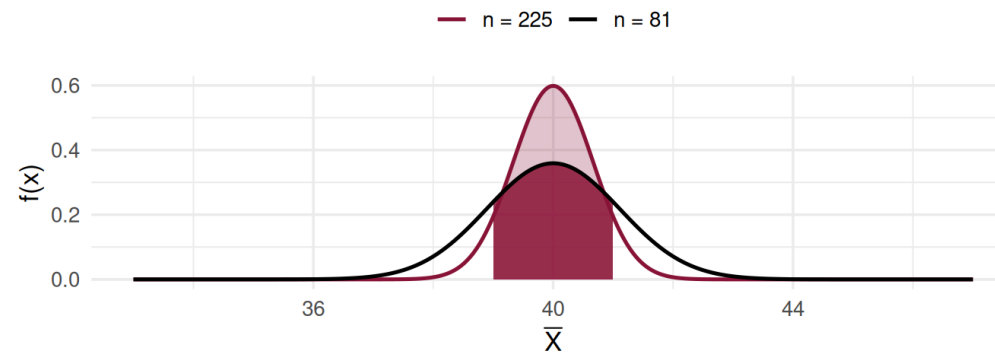
$$\Rightarrow P(39 < \bar{X} \leq 41) \approx P\left(\frac{39 - 40}{\frac{10}{\sqrt{n}}} < Z \leq \frac{41 - 40}{\frac{10}{\sqrt{n}}}\right) =$$

Example 1 — Revisited (cont.)

$$P(39 < \bar{X} \leq 41) \rightarrow (CLT)$$

- $n = 25$: insufficient information for reliable calculations ($n < 30$)
- $n = 81$: $P(-0.9 < Z \leq 0.9) = \Phi(0.9) - \Phi(-0.9) = \mathbf{0.6319}$
- $n = 225$: $P(-1.5 < Z \leq 1.5) = \Phi(1.5) - \Phi(-1.5) = \mathbf{0.8664}$

Effect of Sample Size



Comment: The probability that observations concentrate around the mean increased with the larger sample ($n = 225$).

Example 2

Consider the independent variables:

X_i — number of points on the i -th roll of a die 🎲

Find: $P(2.5 < \bar{X} < 3.5)$ for $n = 1$, $n = 2$, $n = 36$ rolls.

Solution:

$$n = 1: = P(X_1 = 3) = \frac{1}{6}$$

$n = 2$:

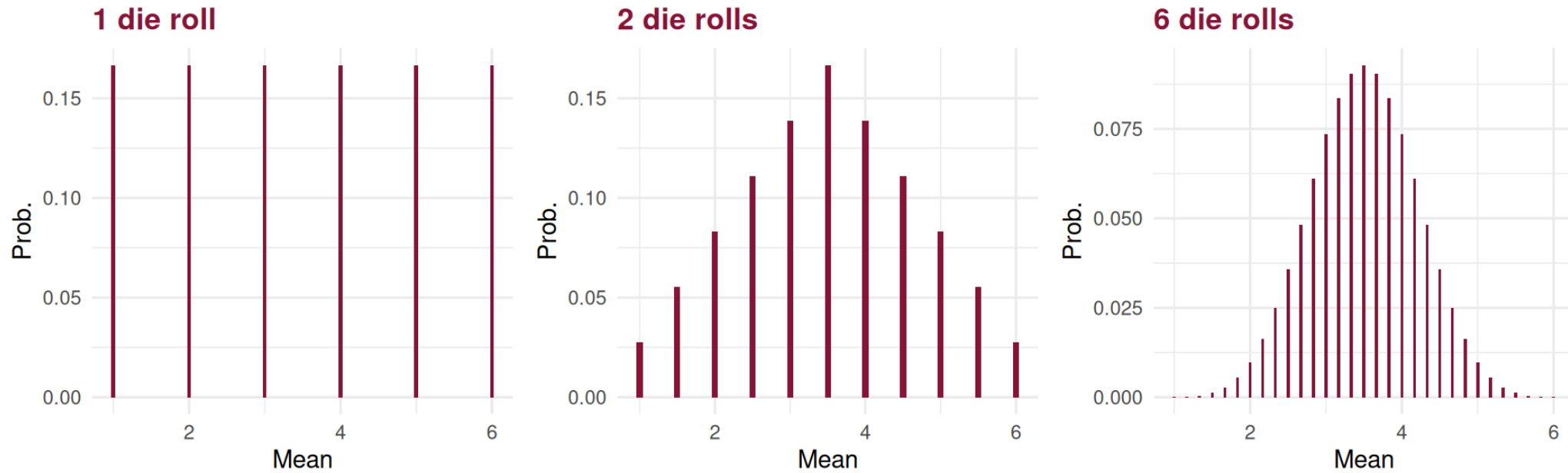
$$\begin{aligned} &= P(X_1 = 1, X_2 = 5) + P(X_1 = 2, X_2 = 4) + P(X_1 = 3, X_2 = 3) \\ &+ P(X_1 = 4, X_2 = 2) + \dots + P(X_1 = 5, X_2 = 1) = \frac{5}{36} \end{aligned}$$

Example 2 — Solution (cont.)

$n = 36$ (CLT): note that for a fair die $\mu = 3.5$ and $\sigma = 1.7078$

$$= P\left(\frac{2.5 - 3.5}{\frac{1.7078}{\sqrt{36}}} < Z < \frac{3.5 - 3.5}{\frac{1.7078}{\sqrt{36}}}\right) = \Phi(0) - \Phi(-3.51) = 0.4998$$

Example 2 — Solution (cont.)



Comments:

1. The plots illustrate convergence towards the Normal distribution.
2. The variable is discrete — see the continuity correction ahead.

Corollaries of the CLT

Corollary 1

Let X be a random variable with a Binomial distribution with parameters n and p , $X \sim \text{binomial}(n, p)$. If $n \geq 30$, $np > 5$ or $n(1 - p) > 5$, then:

$$X \overset{\cdot}{\underset{CLT}{\rightsquigarrow}} \text{Normal}(\mu = np, \sigma = \sqrt{npq}) \Leftrightarrow Z_n = \frac{X - np}{\sqrt{npq}} \overset{\cdot}{\underset{CLT}{\rightsquigarrow}} \text{Normal}(0, 1)$$

Corollary 2

Let X be a random variable with a Poisson distribution with parameter λ , $X \sim \text{Poisson}(\lambda)$. In practice, if $\lambda > 20$, then:

$$X \overset{\cdot}{\underset{CLT}{\rightsquigarrow}} \text{Normal}(\mu = \lambda, \sigma = \sqrt{\lambda}) \Leftrightarrow Z_n = \frac{X - \lambda}{\sqrt{\lambda}} \overset{\cdot}{\underset{CLT}{\rightsquigarrow}} \text{Normal}(0, 1)$$

Continuity Correction

Note

Note: In both corollaries, we are approximating the distribution of a **discrete** random variable with a **continuous** (Normal) distribution. The **continuity correction** should therefore be applied:

$$[F(x)]_{\text{Discrete}} \approx [F(x + 0.5)]_{\text{Normal}}$$

Example: $X \sim \text{Poisson}(\lambda = 30)$

- Exact probability: $P(X = 40) = 0.01394$
- Without correction: $P\left(Z = \frac{40 - 30}{\sqrt{30}}\right) = 0$ — unusable!
- **With continuity correction:**

$$P(X = 40) = P(39.5 \leq X \leq 40.5) \approx P\left(\frac{39.5 - 30}{\sqrt{30}} \leq Z \leq \frac{40.5 - 30}{\sqrt{30}}\right)$$

$$= \Phi(1.92) - \Phi(1.73) = 0.9726 - 0.9582 = \mathbf{0.01444}$$

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Continuity Correction — Other Cases

Remaining cases (discrete X , applying continuity correction):

$$P(X < 40) = P(X \leq 39) = P(X \leq 39.5) \approx \dots$$

$$P(X > 40) = P(X \geq 41) = P(X \geq 40.5) \approx \dots$$

$$P(35 < X < 40) = P(36 \leq X \leq 39) = P(35.5 \leq X \leq 39.5) \approx \dots$$

$$P(35 \leq X < 40) = P(35 \leq X \leq 39) = P(34.5 \leq X \leq 39.5) \approx \dots$$

Example 3

Consider:

1. Independent variables: $X_i \sim \text{binomial}(n_i = 1 ; p = 0.05)$
 $X_i = 1$ if a tax return contains errors (otherwise $X_i = 0$)
2. Consider an inspection of a sample of $n = 1000$ tax returns.

Find the probability that there are **at least 60 incorrect** returns.

Solution: (exact and approximate distributions)

Exact distribution:

$T \sim \text{binomial}(n = 1000 ; p = 0.05)$ — by the additivity of the Binomial

$$P(T \geq 60) = 1 - P(T < 60) = 1 - P(T \leq 59) = \mathbf{0.0867}$$

Example 3 — Approximate Solution

Approximate distribution (exact and approximate):

$$\mu = n_i p = 1 \times 0.05 \quad \sigma = \sqrt{n_i p(1-p)} = \sqrt{1 \times 0.05 \times 0.95} = \sqrt{0.0475}$$

$$T \sim Normal(\mu_T = 1000 \times 0.05 ; \sigma_T = \sqrt{1000} \times \sqrt{0.0475})$$

Applying the continuity correction:

$$P(T \geq 60) \stackrel{\text{cont. corr.}}{=} P(T \geq 59.5) = 1 - P(T < 59.5)$$

$$\stackrel{\text{CLT}}{\approx} 1 - P\left(Z \leq \frac{59.5 - 50}{6.89}\right) = 1 - \Phi(1.38) = \mathbf{0.0838}$$

Comment: The variable is discrete, so we applied a continuity correction.

Example 4 [Exercise 1]

Consider: $T \sim \text{Poisson}(\lambda_T = 30)$

T = number of daily accesses to a website

Find $P(T > 40)$.

Solution: (exact and approximate distributions)

Exact distribution:

$$P(T > 40) = 1 - P(T \leq 40) = 0.0323 \simeq 0.03$$

Note: Since this is a homogeneous Poisson process, the day can be viewed as a sum of n sub-periods, each with the same probability distribution.

Example 4 — Approximate Solution

Approximate distribution:

$$T = X_1 + \dots + X_n \text{ (by additivity), where } X_i \sim \text{Poisson}\left(\lambda = \frac{30}{n}\right)$$

$$T \sim \text{Poisson}(\lambda = 30) \Rightarrow T \overset{\text{CLT}}{\dot{\sim}} \text{Normal}(\mu = \lambda = 30, \sigma = \sqrt{\lambda} = \sqrt{30})$$

Applying the continuity correction:

$$\begin{aligned} P(T > 40) &\underset{\text{cont. corr.}}{=} P(T \geq 40.5) \overset{\text{CLT}}{\approx} P\left(Z \geq \frac{40.5 - 30}{\sqrt{30}}\right) = P(Z \geq 1.92) \\ &= 1 - \Phi(1.92) = 1 - 0.9726 = \mathbf{0.0274} \end{aligned}$$

$$\text{Approximation error: } \varepsilon = |0.0323 - 0.0274| = 0.0049$$